

Deploying OneAgent

Series: ONBRD | **Notebook:** 5 of 10 | **Created:** December 2025 | **Last Updated:** 04/04/2026

Getting Data Into Dynatrace

OneAgent is the foundation of Dynatrace monitoring. This notebook covers deployment strategies, installation methods, and verification steps to ensure your infrastructure is reporting data.

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Prerequisites

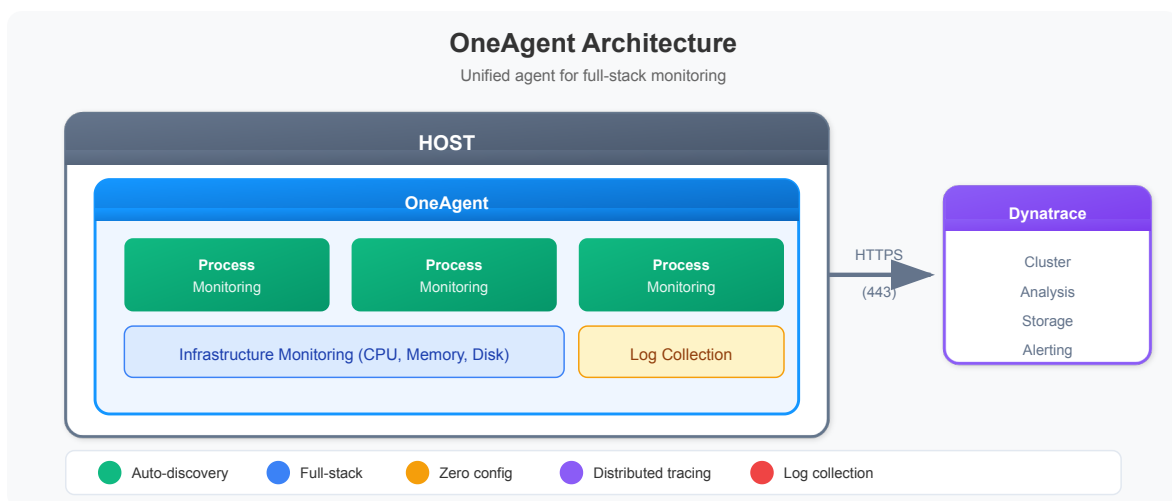
- Admin access to Dynatrace environment
- API token with `InstallerDownload` scope (created in ONBRD-02)
- Root/admin access to target hosts
- Network access from hosts to Dynatrace (port 443) or ActiveGate (ONBRD-03)

- Cloud integrations configured (ONBRD-04) for full context

1. What is OneAgent?

OneAgent is Dynatrace's unified monitoring agent that:

Capability	Description
Auto-discovery	Automatically detects hosts, processes, services
Full-stack	Infrastructure, application, and user experience
Zero configuration	Works immediately after installation
Distributed tracing	End-to-end transaction visibility
Log collection	Collects and forwards log data



2. Deployment Strategy

Phased Rollout (Recommended)

Don't deploy everywhere at once. Use a phased approach:

Phase	Scope	Purpose
Pilot	2-5 non-critical hosts	Validate installation, network, discovery

Phase	Scope	Purpose
Expand	One application tier	Test service detection, tracing
Production	Full environment	Complete coverage

Deployment Methods by Environment

Environment	Recommended Method
Bare metal/VMs	Direct install or package manager
Kubernetes	Dynatrace Operator
OpenShift	Dynatrace Operator
AWS ECS	Task definition sidecar
Azure	VM Extension or AKS Operator
GCP	Direct install or GKE Operator

Network Requirements

OneAgent needs outbound HTTPS (443) to:

- *.apps.dynatrace.com (SaaS)

If direct access isn't possible, OneAgents connect through **ActiveGate** (see ONBRD-03).

3. Generating a Deployment Token

You need an API token with installer download permissions.

Location: Account Management → Access tokens → Generate new token

Required Scope

Scope	API Name	Purpose
PaaS integration - Installer download	InstallerDownload	Download OneAgent installer

Token Naming Convention

Use descriptive names:

- prod-oneagent-deployment
- dev-installer-token
- k8s-operator-token

Important: Copy the token immediately after creation. It won't be shown again.

4. Installation Methods

Linux (Direct Download)

```
# Download the installer
wget -O Dynatrace-OneAgent.sh \
  "https://{tenant-
id}.apps.dynatrace.com/api/v1/deployment/installer/agent/unix/default/latest?
Api-Token={token}&arch=x86&flavor=default"

# Run the installer
sudo /bin/sh Dynatrace-OneAgent.sh
```

Linux (One-liner)

```
wget -O Dynatrace-OneAgent.sh "https://{tenant-
id}.apps.dynatrace.com/api/v1/deployment/installer/agent/unix/default/latest?
Api-Token={token}" && sudo /bin/sh Dynatrace-OneAgent.sh
```

Windows (PowerShell)

```
# Download the installer
Invoke-WebRequest -Uri "https://{tenant-
id}.apps.dynatrace.com/api/v1/deployment/installer/agent/windows/default/late
st?Api-Token={token}" -OutFile Dynatrace-OneAgent.exe

# Run the installer
.\Dynatrace-OneAgent.exe
```

Configuration Management

For Ansible, Puppet, Chef, or other tools, see:

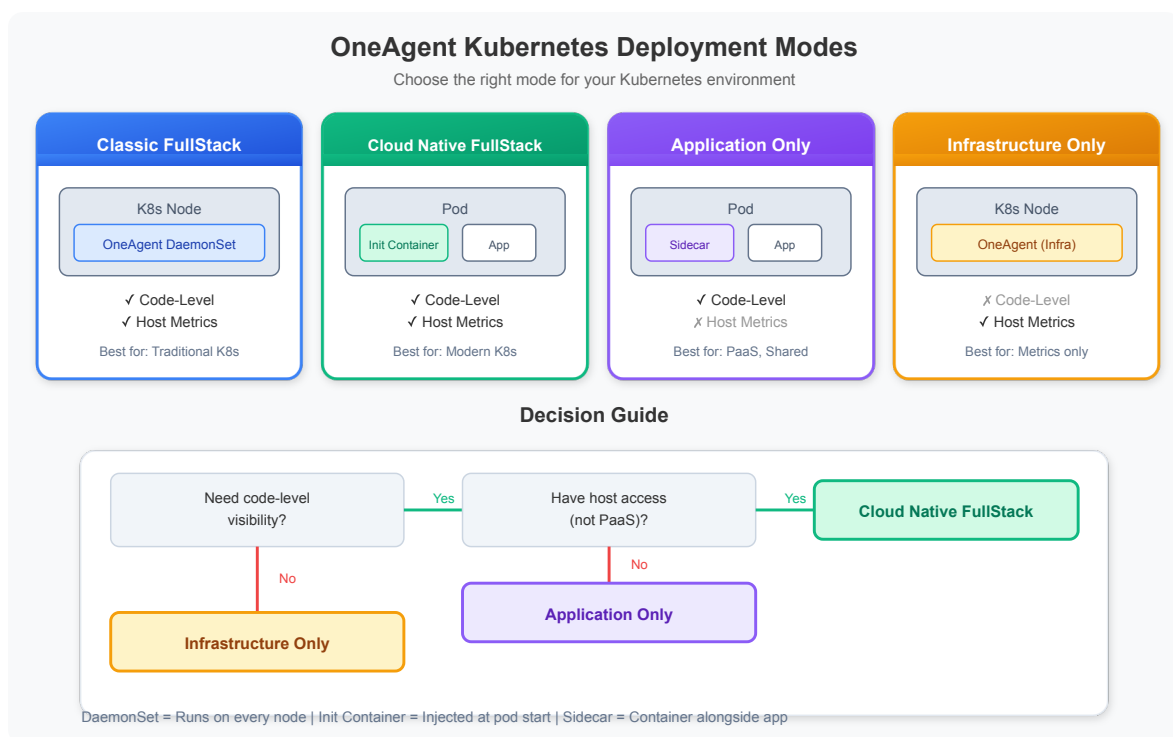
- [Ansible Collection](#)
- [Puppet Module](#)
- [Chef Cookbook](#)

5. Kubernetes Deployment

For Kubernetes environments, use the Dynatrace Operator—the recommended approach for deploying and managing OneAgent in containerized environments.

OneAgent Deployment Modes

The Dynatrace Operator supports multiple deployment modes. Choose based on your requirements:



Mode	Deployment	Code-Level	Host Metrics	Best For
Cloud Native FullStack	Init container + CSI	Yes	Yes	Modern K8s (recommended)

Mode	Deployment	Code-Level	Host Metrics	Best For
Application Only	Sidecar	Yes	No	PaaS, shared nodes, OpenShift
Host Monitoring	DaemonSet	No	Yes	When code-level monitoring not needed

Note: Classic FullStack mode is **not recommended for new deployments**. Dynatrace recommends Cloud Native FullStack for all Kubernetes environments.

Deployment Method: Helm vs Manifests

Method	Pros	Cons	Best For
Helm Charts	Version management, easy upgrades, values override	Requires Helm	Production, GitOps
Manifest Files	Simple, no dependencies	Manual version tracking	Air-gapped, constrained environments

Prerequisites

- `kubectl` access to cluster
- API token with required scopes
- Cluster admin permissions

Required Token Scopes for Kubernetes

Scope	Purpose
<code>InstallerDownload</code>	Download OneAgent
<code>entities.read</code>	Read topology
<code>settings.read</code>	Read configuration
<code>settings.write</code>	Write configuration

Scope	Purpose
DataExport	Export data (optional)

Installation Option 1: Helm (Recommended)

```
# Add Dynatrace Helm repository
helm repo add dynatrace
https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable
helm repo update

# Create namespace and secret
kubectl create namespace dynatrace
kubectl -n dynatrace create secret generic dynakube --from-
literal="apiToken=<API_TOKEN>" --from-
literal="dataIngestToken=<DATA_INGEST_TOKEN>"

# Install with Helm
helm install dynatrace-operator dynatrace/dynatrace-operator \
  --namespace dynatrace \
  --set installCRD=true
```

Installation Option 2: Manifest Files

1. Navigate to Kubernetes monitoring

- Infrastructure → Kubernetes → Add cluster

2. Follow the wizard

- Select your distribution
- Choose observability options
- Download dynakube.yaml

3. Apply to cluster

```
bash kubectl apply -f https://github.com/Dynatrace/dynatrace-
operator/releases/download/v1.8.1/kubernetes.yaml kubectl apply -f
dynakube.yaml
```

4. Verify deployment

```
bash kubectl get pods -n dynatrace
```

DynaKube Custom Resource Configuration

The DynaKube CR is the primary configuration for the Dynatrace Operator. Here are examples for each deployment mode:

Important: Use `apiVersion: dynatrace.com/v1beta5` or `v1beta6` for Dynatrace Operator 1.8.x. Operator 1.8.0 removes `v1beta3` and auto-converts to `v1beta6`. `v1beta6` adds OTLP exporter configuration.

Cloud Native FullStack (Recommended for most K8s):

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # Cloud Native FullStack – automatic init container injection
  oneAgent:
    cloudNativeFullStack:
      tolerations:
        - effect: NoSchedule
          key: node-role.kubernetes.io/master
          operator: Exists
      args:
        - --set-host-group=my-k8s-cluster

  # ActiveGate for Kubernetes API monitoring
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
      - dynatrace-api
```

Application Only (for PaaS or shared nodes):


```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # Application Only – sidecar injection, no host monitoring
  oneAgent:
    applicationMonitoring:
      useCSIDriver: true
```

Host Monitoring (host metrics only):

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # Host monitoring only – no code-level visibility
  oneAgent:
    hostMonitoring: {}
```

Air-Gapped / Private Registry Deployment

For environments without internet access, host images in a private registry:

```

apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # Custom image locations for air-gapped environments
  oneAgent:
    cloudNativeFullStack:
      image: my-registry.example.com/dynatrace/oneagent:latest

  activeGate:
    capabilities:
      - kubernetes-monitoring
    image: my-registry.example.com/dynatrace/activegate:latest

```

Image Mirroring Script:

```

# Mirror required images to private registry
IMAGES=(
  "docker.io/dynatrace/dynatrace-operator:v1.8.1"
  "docker.io/dynatrace/dynatrace-oneagent:latest"
  "docker.io/dynatrace/dynatrace-activegate:latest"
)

for img in "${IMAGES[@]"; do
  docker pull $img
  docker tag $img my-registry.example.com/${img#*/}
  docker push my-registry.example.com/${img#*/}
done

```

Kubernetes Metadata and Dynatrace Tags

Dynatrace can leverage Kubernetes labels and annotations for observability:

Feature	Behavior	Limitations
Pod labels	Appear as <code>[Kubernetes]</code> context tags on processes	Requires OneAgent code module injection
Annotations	Available as custom metadata at Process level	Requires view RBAC permissions

Feature	Behavior	Limitations
Namespace labels	Available via Primary Grail tags	Does not enrich K8s metrics or events

Requirements:

- Service accounts need `view` access via rolebinding/clusterrolebinding
- Labels are read via Kubernetes REST API at deployment time
- Not all entities receive automatic tags (namespaces, pods, workloads have limited support)

For more control, configure automatic tagging rules: **Settings → Automatically applied tags**

Multi-Tenant / Namespace Isolation

For large clusters with multiple teams, use namespace selectors:

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: team-a-dynakube
  namespace: dynatrace
spec:
  apiUrl: https://team-a-tenant.live.dynatrace.com/api

  oneAgent:
    cloudNativeFullStack:
      namespaceSelector:
        matchLabels:
          dynatrace-tenant: team-a
```

This allows different teams/tenants to monitor different namespaces within the same cluster.

6. Verifying Deployment

After installation, verify OneAgent is reporting data.

```
// Check all hosts with OneAgent
fetch dt.entity.host
| fields entity.name, state, monitoringMode
| filter state == "RUNNING"
| sort entity.name
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | fields name, state, monitoringMode
// | filter state == "RUNNING"
// | sort name
// | limit 50
```

```
// Check hosts by monitoring state
fetch dt.entity.host
| summarize host_count = count(), by: {state, monitoringMode}
| sort host_count desc
```

```
// Check hosts by monitoring state – useful for verifying deployment
fetch dt.entity.host
| fields entity.name, state, monitoringMode
| summarize host_count = count(), by: {state}
| sort host_count desc
```

```
// Check discovered services
fetch dt.entity.service
| fields entity.name, serviceType
| summarize service_count = count(), by: {serviceType}
| sort service_count desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | fields name, serviceType
// | summarize service_count = count(), by: {serviceType}
// | sort service_count desc
```

```
// Check for processes discovered
fetch dt.entity.process_group
| fields entity.name
| sort entity.name
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes PROCESS
// | fields name
// | sort name
// | limit 50
```

Host Verification Commands

Linux:

```
# Check OneAgent status
sudo systemctl status oneagent

# Check connection to Dynatrace
sudo /opt/dynatrace/oneagent/agent/tools/oneagent-connection-check

# Check OneAgent logs
sudo tail -100 /var/log/dynatrace/oneagent/oneagent.log
```

Windows:

```
# Check service status
Get-Service -Name "Dynatrace OneAgent"

# Check connection
& "C:\Program Files\dynatrace\oneagent\agent\tools\oneagent-connection-check.exe"
```

7. Troubleshooting

Common Issues

Issue	Cause	Solution
Host not appearing	Network blocked	Check firewall rules for 443
Host showing but no data	Agent not running	Restart OneAgent service
Services not discovered	Processes not restarted	Restart monitored applications
Old OneAgent version	Auto-update disabled	Enable auto-update or manual upgrade
Connection errors	Proxy required	Configure ActiveGate or proxy

Network Verification

```
# Test connectivity to Dynatrace
curl -v https://{tenant-id}.apps.dynatrace.com/api/v1/time

# Check DNS resolution
nslookup {tenant-id}.apps.dynatrace.com

# Test with OpenSSL
openssl s_client -connect {tenant-id}.apps.dynatrace.com:443
```

Process Restart Requirement

OneAgent injects into running processes. After initial installation:

- **New processes** - Automatically monitored
- **Existing processes** - Restart required for full monitoring

For deep code-level visibility, restart:

- Application servers (Tomcat, JBoss, etc.)
- Web servers (Apache, Nginx, IIS)
- .NET/Java applications
- Node.js applications

8. Next Steps

With OneAgent deployed and verified:

1. **ONBRD-06: Organizing Your Environment** - Set up tags, segments, and naming conventions
2. Wait 15-30 minutes for full topology discovery
3. Check Smartscape for service dependencies
4. Plan broader rollout based on pilot results

Deployment Checklist

- ☐ Pilot hosts deployed (2-5 hosts)
 - ☐ Hosts appearing in Dynatrace
 - ☐ OneAgent status "RUNNING" on all hosts
 - ☐ Services being discovered
 - ☐ Connection check passing
 - ☐ Smartscape showing topology
-

Summary

In this notebook, you learned:

- What OneAgent does and how it works
 - Phased deployment strategy
 - How to generate deployment tokens
 - Installation methods for Linux, Windows, and Kubernetes
 - How to verify successful deployment
 - Common troubleshooting steps
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References

- [OneAgent Overview](#)
 - [Linux Installation](#)
 - [Windows Installation](#)
 - [Kubernetes Operator](#)
 - [Deployment API](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.